Vaibhav Saini

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SUMMARY

Early career Software Engineer focused on Applied AI, Generative AI, and full-stack software development, with hands-on experience building production oriented RAG systems, LLM inference services, model evaluation pipelines, AI routing systems, Python backends, and TypeScript applications. Experienced with Python, FastAPI, React, TypeScript, Docker, AWS, Redis, REST APIs, structured logging, RBAC, and scalable backend services. Built AI systems spanning hybrid retrieval, model inference, evaluation, verification, observability, and cloud-integrated application development. M.S. in Information Technology with a Security focus from Arizona State University.

EXPERIENCE

Software Engineer
Badoliasoft | June 2023 - July 2024

- Implemented JWT authentication with HMAC-SHA512 hashing, 3 tier RBAC, token blacklisting, and IP-based rate limiting for a fintech platform supporting 100+ users, with zero unauthorized access incidents post deployment.
- Built real time application workflows using SignalR with RabbitMQ fanout messaging and integrated Azure Blob Storage with CDN, reducing average file delivery latency by 40%.
- Implemented Redis distributed caching, structured logging, and global exception handling middleware, reducing database load by 35% while improving backend reliability and production observability.
- Contributed to a full-stack Angular + .NET 8 platform spanning 200+ C# files and 10 REST controllers, using TypeScript, JavaScript, REST APIs, and cloud services across user management, messaging, and project workflows.
- Operated as a core contributor on a small engineering team building a greenfield fintech platform from concept through delivery, owning authentication, notifications, caching, debugging, and production oriented backend systems.

SELECTED PROJECTS

HomHive | Agentic AI Household Intelligence Platform | NVIDIA × Nebius AI Hackathon | Sep 2026 - Present | Live Demo | GitHub

- Architected and deployed an agentic household-intelligence platform using Python, FastAPI, PostgreSQL, SQLAlchemy, Alembic, Next.js, Docker, Vercel, and Railway, transforming multimodal household evidence into persistent temporal state, reactive and predictive tasks, forecasts, and constraint-aware action plans, supported by 348 passing backend tests.
- Engineered a multimodal perception pipeline on Nebius with evidence grounded validation and detection-first scene understanding, increasing sceneentity recall from 47.5% to 71.2% (+50% relative) and micro-F1 from 58.3% to 71.8% (+13.5 percentage points) across a human-labeled 59-entity development evaluation set.
- Integrated NVIDIA Nemotron reasoning and Tavily research to generate and dynamically replan household actions across time availability, presence, quiet hours, priority, and feasibility constraints, demonstrating live context-sensitive replanning from 20 to 5 available minutes across a production Vercel - Railway - PostgreSQL - Nebius/Nemotron deployment.

SecRAG Pro | Agentic RAG & Cost-Optimized GenAI Platform | Dec 2025 - Mar 2026 | GitHub

- Designed and built a production-oriented agentic RAG backend integrating hybrid BM25 + dense retrieval, Reciprocal Rank Fusion, cross encoder reranking, multi provider LLM routing, evaluation pipelines, and REST APIs, supporting 513 evaluation requests with 0 failures across retrieval, reasoning, and answer-generation workflows.
- Engineered an adaptive LLM cost routing layer across GPT-4o, GPT-4o Mini, Claude Sonnet, Claude Haiku, and local Llama 3.2 via Ollama, achieving 62.3% lower inference cost vs. an all-GPT-4o baseline while maintaining 4.15/5 answer quality and 97%+ query-complexity classification accuracy.
- Implemented grounded answer verification and adaptive retrieval, using LLM-as-judge claim validation, citation checks, variable top-k retrieval, and intent-aware context selection, achieving 99.6% citation accuracy on a 50-question evaluation set while reducing unsupported responses and improving retrieval coverage across distributed document sections.

EDUCATION

Master of Science in Information Technology (Security) | Arizona State University | Tempe, AZ - 2026 | **3.97 / 4.0**
Bachelor of Engineering in Computer Science (Data Science) | MMDU | Ambala, Haryana, India - 2023 | **7.62 / 10.0**

CERTIFICATIONS

Machine Learning Specialization | DeepLearning.AI - 2026
Generative Al and LLMs | DeepLearning.AI - 2026
AWS Academy Cloud Architect and Solutions Engineer | Amazon Web Services - 2025
AWS Certified Cloud Practitioner | Amazon Web Services - 2025

SKILLS

- **Languages & Frameworks:** Python, C#, Java, TypeScript, JavaScript, SQL, FastAPI, ASP.NET Core, Next.js, React, Angular, Node.js, REST APIs, GraphQL
- **AI & ML:** Generative AI, Agentic AI, Multimodal AI, LLMs, RAG, LLM Inference, LLM Routing, Hybrid Retrieval, BM25, Dense Vector Retrieval, Reciprocal Rank Fusion, Cross-Encoder Reranking, LLM-as-Judge, Model Evaluation, Evaluation Pipelines, NVIDIA Nemotron, Nebius AI, Tavily, Ollama
- **Cloud, Data & Infrastructure:** AWS, Azure, PostgreSQL, AWS RDS, SQLAlchemy, Alembic, Redis, Docker, Railway, Vercel, RabbitMQ, SignalR
- **Engineering:** Distributed Systems, API Design, Object-Oriented Design, Data Structures & Algorithms, JWT, RBAC, Testing, Debugging, Observability, Structured Logging, CI/CD, Git, GitHub